Dimensionality Reduction and PCA

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Content

- **PCA (Principal Component Analysis)**
  - A **dimensionality reduction** technique, high-dimensional data to low dimensions
  - Used for **data visualization, noise reduction, and feature extraction**
- **t-SNE (t-Distributed Stochastic Neighbor Embedding)**
  - A **non-linear dimensionality reduction** method
  - Excellent for **visualizing high-dimensional data** in 2D or 3D
- **Isolation Forest**
  - An **unsupervised anomaly detection** algorithm
- **Statistical Outlier Method**
  - Identifies outliers using **statistical measures**
  - Common approaches: **Z-score, IQR (Interquartile Range)**
- **Fraud Heatmap**
  - A visual analytics tool, Helps identify: High-risk regions, Time-based fraud trends,
  - Often used with anomaly detection outputs

Principal Component Analysis (PCA)

- A **dimensionality reduction** technique
  - Converts high-dimensional data into fewer **principal components**

- Used for **data visualization, noise reduction, and feature extraction**
  - **Projection of N-Dim data to 2D or 3D**
- Preserves **maximum variance**

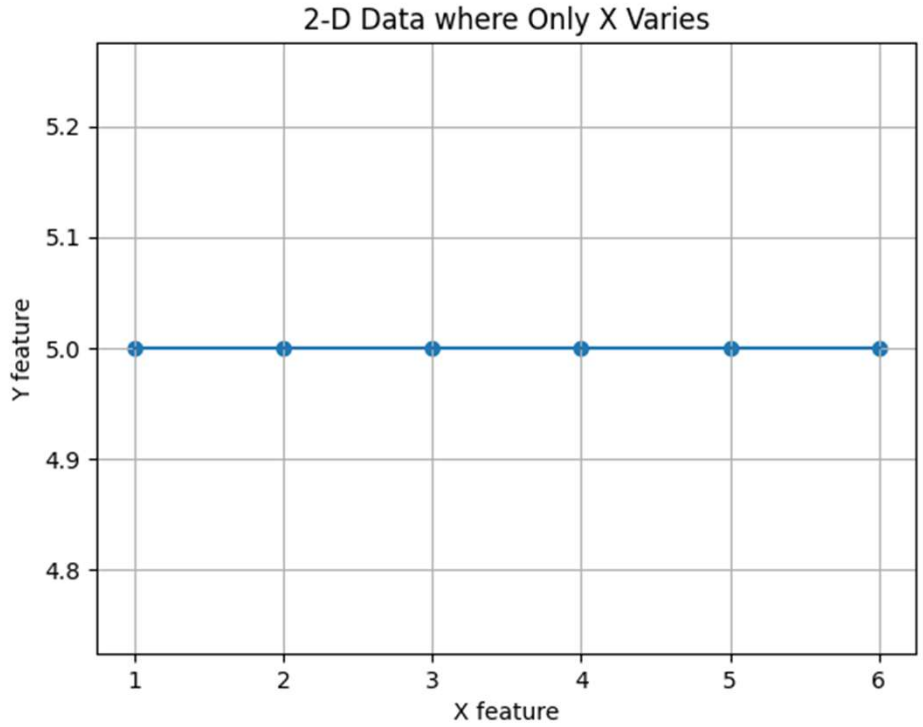
Applications

- **Data Visualization**:
  - When you have data with many features (dimensions), it's impossible to visualize. PCA reduces it to 2D or 3D while preserving the most important patterns.
- **Noise Reduction**:
  - By keeping only the principal components with the most variance, you filter out noise and keep the signal.
- **Feature Extraction**:
  - PCA creates new features that are combinations of original features, capturing the most important information in fewer dimensions.
- **Computational Efficiency**:
  - Fewer dimensions mean faster training for machine learning models and less storage space.
- **Multicollinearity Removal**:
  - PCA creates uncorrelated components, which helps with models sensitive to correlated features.

PCA - Visualization Intuition

- Example 1: 2-D Dataset representing objects using two features

Point	Feature X	Feature Y
1	1	5
2	2	5
3	3	5
4	4	5
5	5	5
6	6	5



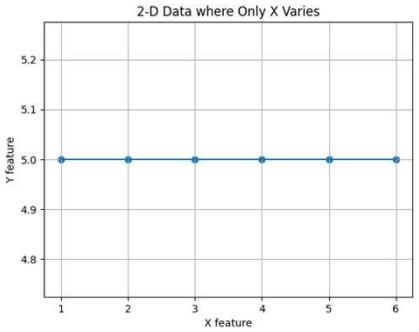
For **data representation**, do we need two features two discriminate objects? No

PCA - Visualization Intuition

- For **data representation**, do we need two features to discriminate objects? No
- Why?
  - Data variations are present only with respect to X feature
  - Only this feature is helpful in differentiating the objects
  - So, if the objects represented only one feature (which is X), the objects will be equally differentiable
  - So 2-D data can be just represented in 1-D

Point	Feature X	Feature Y
1	1	5
2	2	5
3	3	5
4	4	5
5	5	5
6	6	5

representing objects using two features (2-D data)



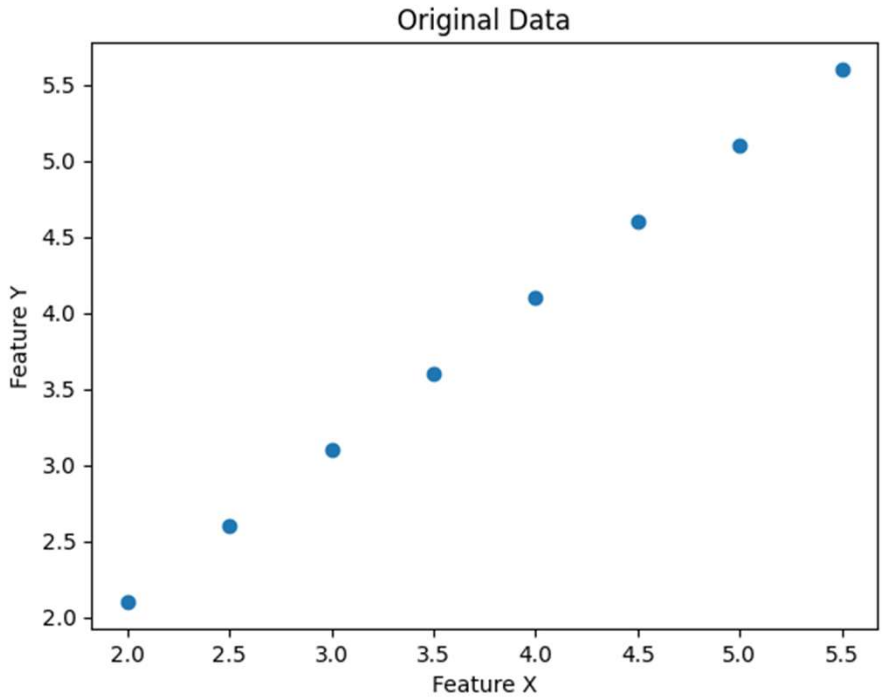
Point	Feature X
1	1
2	2
3	3
4	4
5	5
6	6

representing objects using 1 features (1-D data)

PCA - Visualization Intuition

- Example 2: 2-D Dataset representing objects using two features

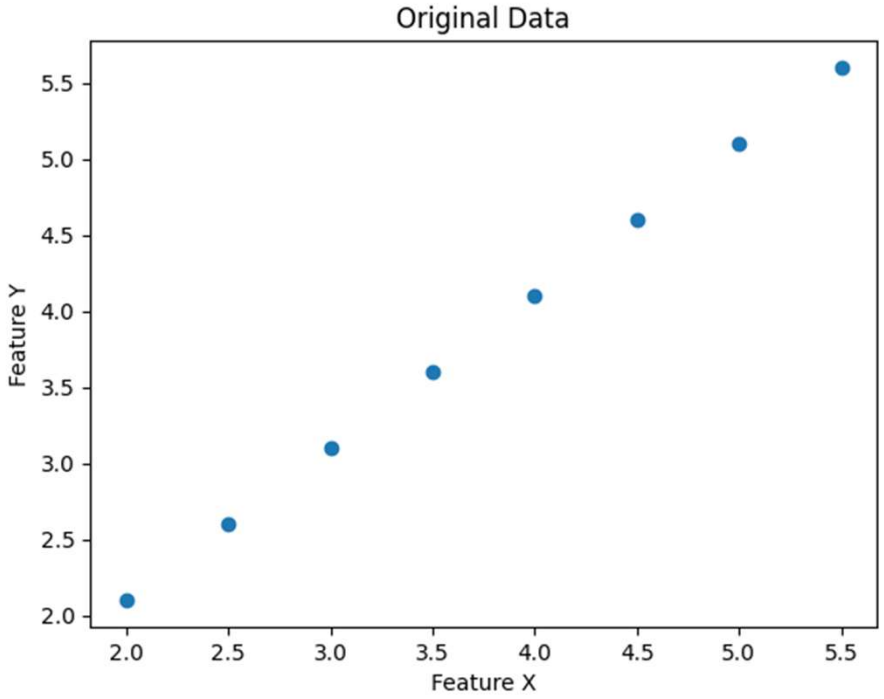
Point	Feature X	Feature Y
1	2.0	2.1
2	2.5	2.6
3	3.0	3.1
4	3.5	3.6
5	4.0	4.1
6	4.5	4.6
7	5.0	5.1
8	5.5	5.6

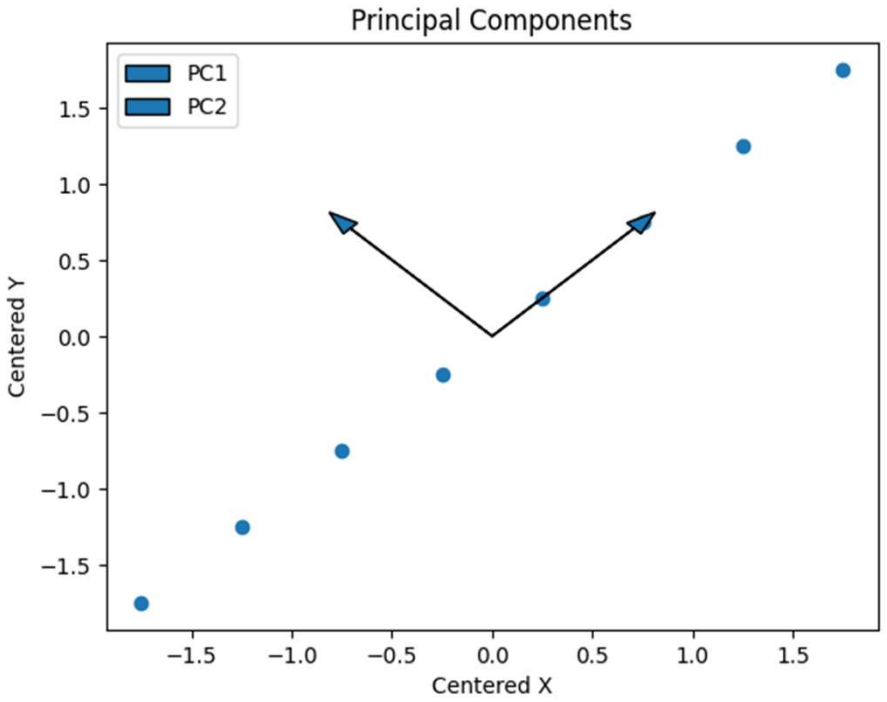


For **data representation**, do we need two features two discriminate objects? No

What PCA does with the data (Core Idea)

- **PCA** find directions where the data varies the most, and then **projects the data onto those directions**.

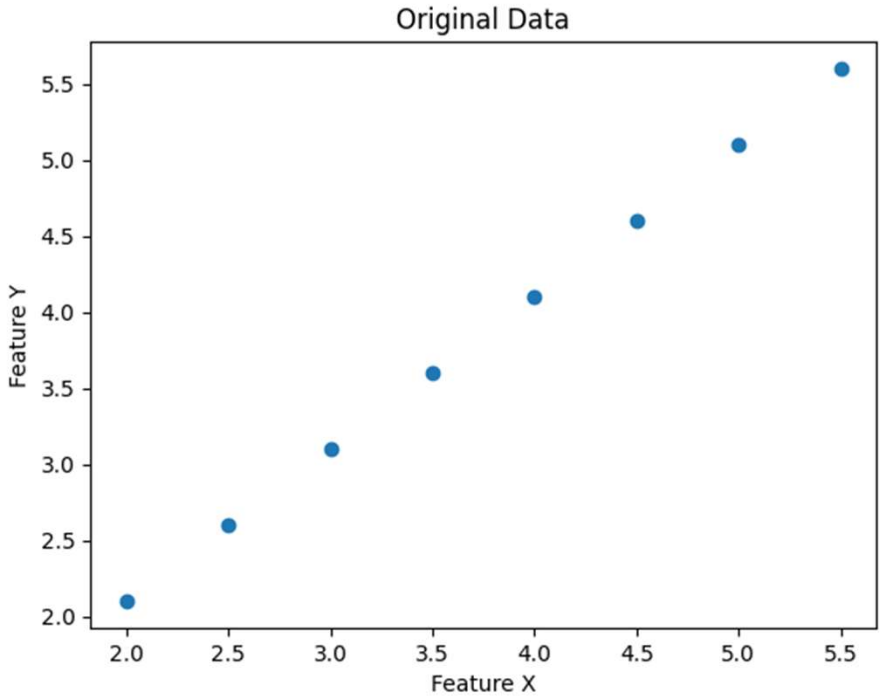




PCA - Steps

- **Given data matrix**

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{bmatrix}$$



PCA - Steps

- Given data matrix

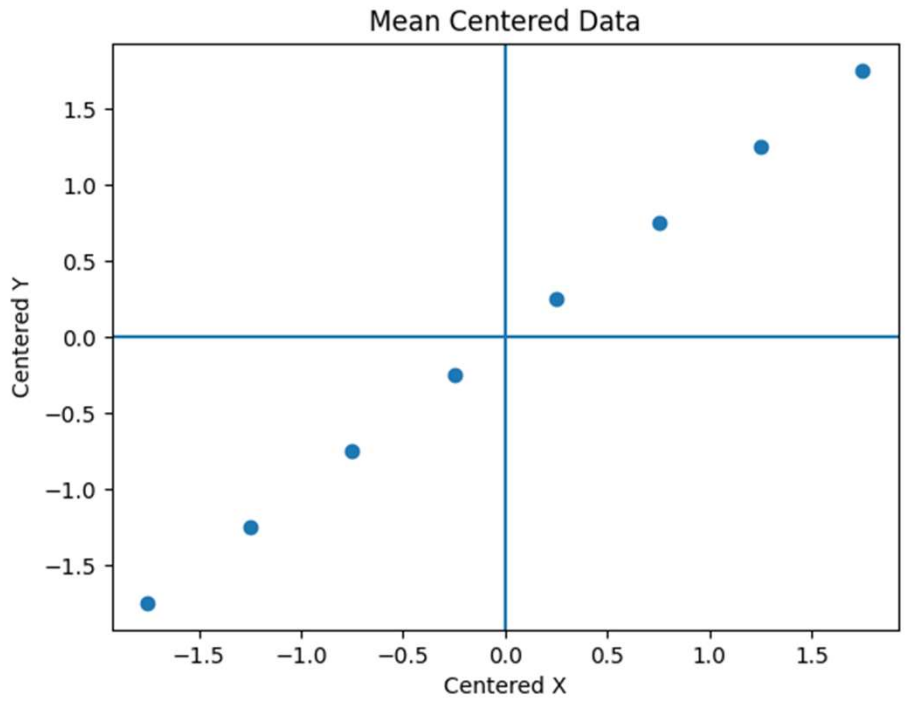
$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{bmatrix}$$

Compute mean of each feature

$$\mu = [\mu_1 \quad \mu_2], \textit{where} \quad \mu_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$$

Center the data

$$X_c = X - \mu$$



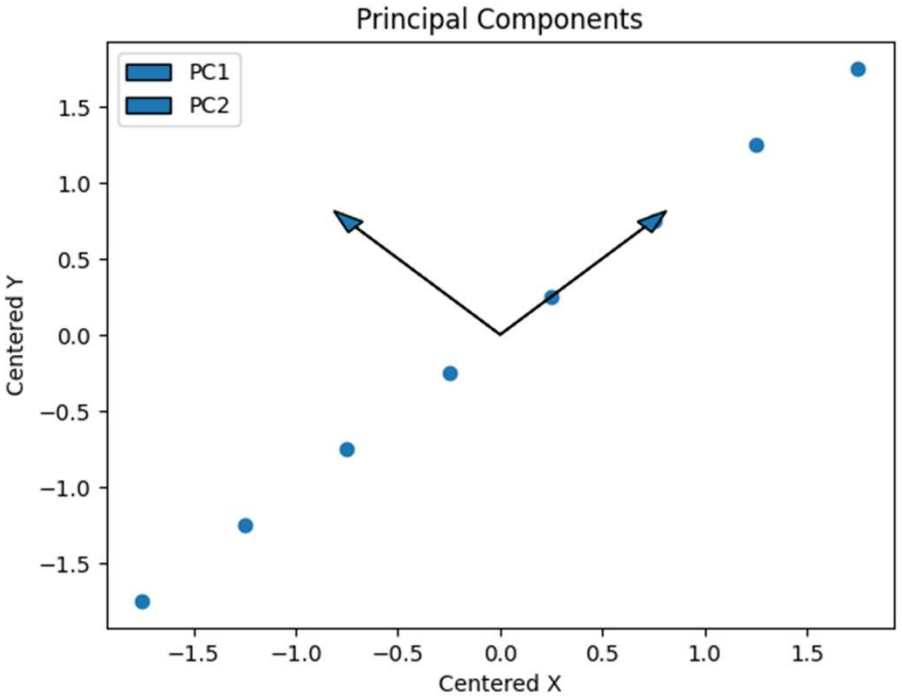
PCA - Steps

- **Eigenvalues and Eigenvectors**

Solve:

$$\Sigma v = \lambda v$$

- Where:
  - v= eigenvector (direction)
  - λ= eigenvalue (variance along that direction)
- For 2-D:
  - 2 eigenvalues
  - 2 orthogonal eigenvectors



PCA - Steps

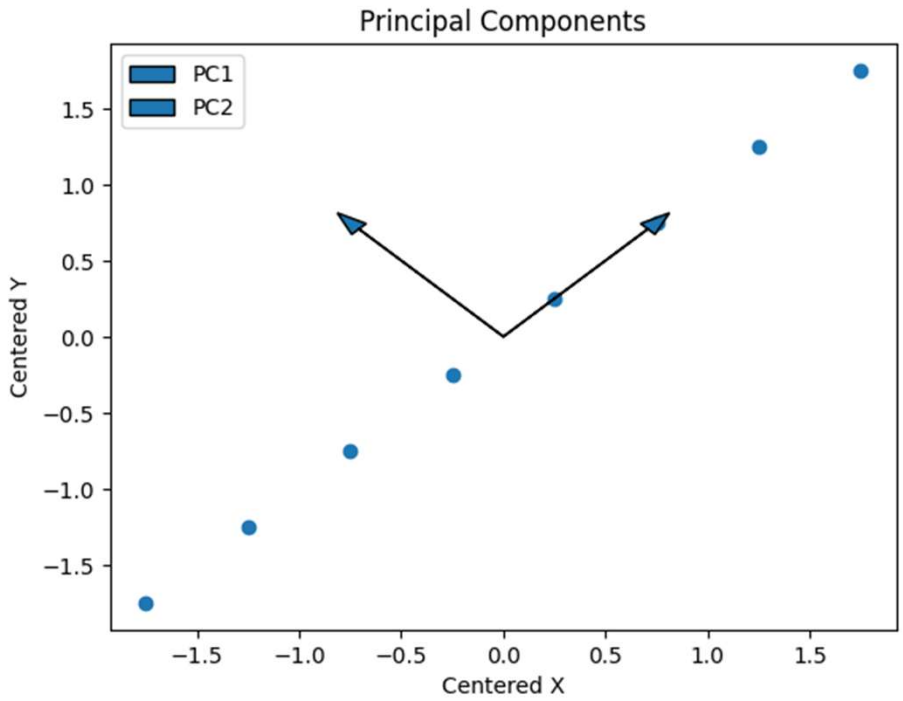
- **Select the Top Principal Component**
  - Sort eigenvalues:

$$\lambda_1 \geq \lambda_2$$

  - Choose eigenvector v_1 corresponding to largest eigenvalue λ_1
  - **Why?**

$$\lambda_1 = \max_{\|v\|=1} Var(X_c v)$$

  - This direction captures **maximum variance**

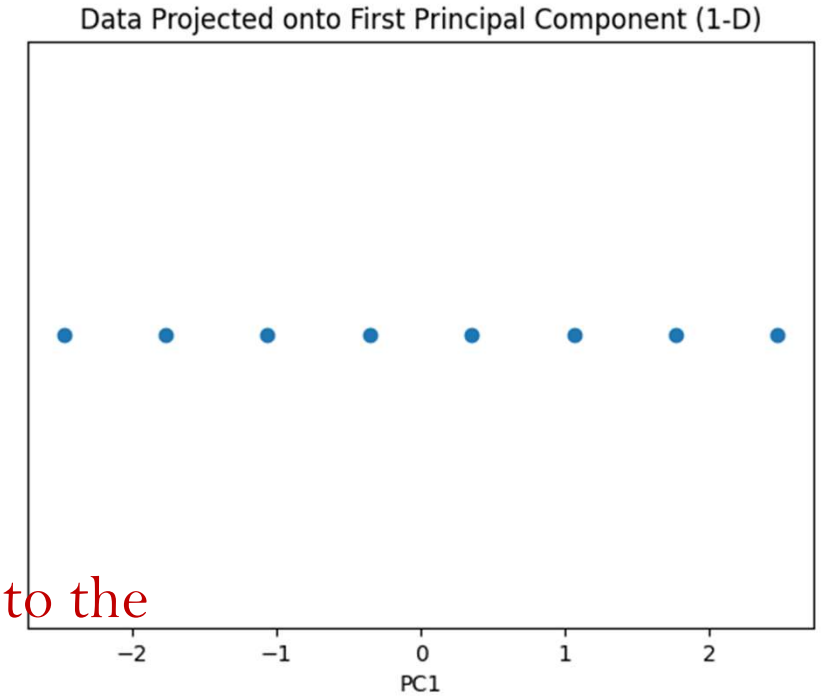


PCA - Steps

- **Project Data onto Principal Component**
  - **Projection formula:** $Z = X_c v_1$
  - Where:
  - Z= 1-D representation
  - Each value is a **dot product**

$$z_i = x_i \cdot v_1$$

- **Geometric meaning**
  - Drop a perpendicular from each point onto the PC axis.



Summary Table

Step	Math	Meaning
Center	$X - \mu$	Remove bias
Covariance	$\frac{1}{n-1} X^T X$	Measures spread
Eigen	$\Sigma v = \lambda v$	Find directions
Select PC	max λ	Keep most info
Project	Xv	Reduce dimension

Thank You